

Lab Report Questions

Instructions:

- Create your new repository on GitHub and push your lab solution to GitHub with new branch for each lab set.
- Share your repository to instructor.basanta@gmail.com as a collaborator.
- Print and attach your updated GitHub screenshot into lab set

Format

- Cover (Print)
- **GitHub Screenshot**
- Lab Question –hand written
- Introduction –hand written
- Syntax and Format –hand written
- Code –hand written
- Output(Print)- With full URL

Lab-1

1. Write a JavaScript function that reverse a number.

2. Write a JavaScript function that returns a passed string with letters in alphabetical order.

Example string: 'webmaster'
Expected Output: 'abeemrstw'

3. Write a JavaScript function that accepts a string as a parameter and counts the number of vowels within the string.

4. Write a JavaScript function that accepts a string as a parameter and converts the first letter of each word of the string in upper case.

5. Write a JavaScript program to get the current date. **Expected Output** : mm-dd-yyyy, mm/dd/yyyy or dd-mm-yyyy, dd/mm/yyyy, yyyy month date day

6. Write a JavaScript program to calculate number of days left until next Christmas.

7. Write a JavaScript program to calculate addition, subtraction, multiplication and division of two numbers (input from user).

8. Write a JavaScript program that accept two integers and display the larger and smaller.

9. Write a JavaScript program that accept three integers and display the larger and smaller.

10. Write a JavaScript program that accept n number of integers into array and display the larger.

11. Write a JavaScript program that accept n number of integers into array and display the smaller.

12. Write a JavaScript function to check whether an `input` is an array or not

13. Write a simple JavaScript program to join all elements of the following array into a string

14. Write a JavaScript program to sort the items of an array.

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Sample array : `var arr1 = [ 3, 8, 7, 6, 5, -4, 3, 2, 1 ]`; Sample Output : -4,-3,1,2,3,5,6,7,8

15. Write a JavaScript program to display the colors in the following way : Here is the sample array: color = ["Blue ", "Green", "Red", "Orange", "Violet", "Indigo", "Yellow "]; o = ["th", "st", "nd", "rd"]

Output

"1st choice is Blue ."

"2nd choice is Green."

"3rd choice is Red."

16. Write a JavaScript for loop that will iterate from 0 to 15. For each iteration, it will check if the current number is odd or even, and display a message to the screen.

17. Write a JavaScript conditional statement to find the largest of five numbers. Display an alert box to show the result taking input using prompt

18. Write a JavaScript function to get difference between two dates in days.

19. Write a JavaScript program to calculate age taking input using form.

20. Write a JavaScript function to get time differences in days between two dates.

21. Write a JavaScript function to get the weekend date.

22. Write a JavaScript function to check whether a string is blank or not.

23. Write a JavaScript function to remove specified number of characters from a string

24. Write a JavaScript program to perform following task.
- Create an empty object called dog
 - Print the the dog object on the console
 - Add name, legs, color, age and bark properties for the dog object. The bark property is a method which return woof woof
 - Get name, legs, color, age and bark value from the dog object
 - Set new properties the dog object: breed, getDogInfo

25. Create an object literal called personAccount. It has firstName, lastName, incomes, expenses properties and it has totalIncome, totalExpense, accountInfo, addIncome, addExpense and accountBalance methods. Incomes is a set of incomes and its description and expenses is a set of incomes and its description.

26. Write a function called *tenMostFrequentWords* which get the ten most frequent word from a string taking user input from user?

27. Write JavaScript program to perform following steps of task for DOM manipulation a. Create an index.html file and put four p elements as above: Get the first paragraph by using document.querySelector(tagname) and tag name

- b. Get each of the the paragraph using document.querySelector('#id') and by their id
- c. Get all the p as nodeList using document.querySelectorAll(tagname) and by their tag name
- d. Loop through the nodeList and get the text content of each paragraph e. Set a text content to paragraph the fourth paragraph,Fourth Paragraph f. Set id and class attribute for all the paragraphs using different attribute setting methods
- g. Change stye of each paragraph using JavaScript(eg. color, background, border, font-size, font-family)
- h. Select all paragraphs and loop through each elements and give the first and third paragraph a color of green, and the second and the fourth paragraph a red color
- i. Set text content, id and class to each paragraph

28. Develop the following application; use the following HTML elements to get started with. You will get the same code on starter folder. Apply all the styles and functionality using JavaScript only.

- a. The year color is changing every 1 second
- b. The date and time background color is changing every on seconds
- c. Completed challenge has background green
- d. Ongoing challenge has background yellow
- e. Coming challenges have background red

```
<div class="wrapper">
  <h1>Asabeneh Yetayeh challenges in 2020</h1>
  <h2>30DaysOfJavaScript Challenge</h2>
  <ul>
    <li>Python Challenge Done</li>
    <li>JavaScript Challenge Ongoing</li>
    <li>React Challenge Coming</li>
    <li>FullStack Challenge Coming</li>
    <li>DataAnalysis Challenge Coming</li>
    <li>ReactNative Challenge Coming</li>
    <li>MachineLearning Challenge Coming</li>
  </ul>
</div>
```

29. Create a div container on HTML document and create 0 to 100 numbers dynamically and append to the container div.

- a. Even numbers background is green
- b. Odd numbers background is yellow
- c. Prime numbers background is red

30. Write JavaScript program that use countries array to display 50 countries into tabular format 6 countries in a row.

31. Write JavaScript program that generate numbers from 1 to user input number and marking evens, odds and prime numbers with three different colors

32. Write JavaScript program to generate following program counter



33. Write JavaScript program to create following calculator and calculate result accordingly

1			c
1	2	3	/
4	5	6	*
7	8	9	-
0	.	=	+

34. Create analog clock using HTML, CSS and JavaScript



35. Create a digital clock using HTML, CSS and JavaScript.



36. Write a JavaScript Program to perform following task.

- a. to accept Key and Values from user and store into cookie
- b. to read cookie according to key entered from user
- c. to display all cookies into tabular format with **key : values**
- d. to delete cookie according to key entered from user
- e. delete all the cookies

37. Write JavaScript program to clone element and remove element from table a. If user click on add new button new row must be added as a last row b. If user click on trash icon, remove related row from DOM

38. Write JavaScript program with different button and paragraph and perform following event according to button pressed

Change Background	Change Paragraph Background	Hide Paragraph	Show Paragraph	Increase font Size by 2pt.	Decrease font Size by 2pt.	Reset font into 12 pt.	Change Text Color
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39. Design and validate form with following condition using HTML, CSS and JavaScript. a. Form field: Name, Address, Username, Email, Password, Website, Phone, Gender(Radio), Course(Dropdown),

b. Name Validation: We check to make sure the name field is not empty and does not contain any numbers.

c. Username Validation: We check to make sure the username field is not empty and does not contain any spaces and special character except underscore (_).

d. Phone Validation: We check to make sure the phone field is not empty, does not contain any character except numbers, and should start with [98/97/96].

- e. Gender Validation: We check the gender field is not empty.
- f. Address Validation: We check that the address field is not empty.
- g. Email Validation: We make sure that the email field is not empty and that it includes the “@” symbol.
- h. Password Validation: We ensure that the password field is not empty and that the password is at least eight characters long with at least one digit, one upper case character, one lowercase character and one special character.
- i. Course Selection Validation: We check that a course has been selected from a dropdown list.